

1 Dinamical Systems

1.1 Models of Biological Systems

Logistic Growth Model (Population): Model: $\frac{dN}{dt} = rN(1 - \frac{N}{K})$ N : population at time t ; K : carrying capacity (depends on resources); $r, K > 0$.
Sol: $N(t) = \frac{N_0 K e^{rt}}{K + N_0(e^{rt} - 1)} \rightarrow K$ **Equilibrium points:** $N = 0$ (unstable), $N = K$ (stable). **Quickest:** $N = \frac{K}{2}$ ($<$: 凹; $>$: 凸)

Delay Model: Model: $\frac{dN}{dt} = f(N(t), N(t-T))$ T : delay time > 0 周期解 | 与初始值无关 ps: To consider e.g. gestation period, maturation time etc.

¹Ext. of Logistic Model: $\frac{dN}{dt} = rN(t) [1 - \frac{N(t-T)}{K}]$ (用"ave. pop." 估计, 准用"Convolution Type") $\Rightarrow$ ²**Dimensionless:** $\frac{dN^*}{dt^*} = N^*(t^*) [1 - N^*(t^* - T^*)]$ ($N^* = \frac{N}{K}$, $t^* = rt$, $T^* = rT$)
^{1,2} **Periodic sol:** stable limit cycle $N(t + t_p) = N(t)$, t_p : period. **Amplitude:** rT **Equilibrium Points:** $N = 0 | N^* = 0$ (unstable); $N = K | N^* = 1$ (Stable: $0 < T^* = rT < \frac{\pi}{2}$; Unstable: $T^* > \frac{\pi}{2}$)

Continuous Population Model (EN): Model: $\frac{dN}{dt} = rN(1 - \frac{N}{K}) - EN = f(N)$ $r, K, E > 0$; EN : harvesting rate/unit time; E : effort measure.

Equilibrium: $N = 0$ (unstable); $N = N_h(E) = K(1 - \frac{E}{r}) > 0$ (need $E < r$, Stable) **Yield:** $Y(E) = EN \cdot E > r$: die out; else: **Max Yield:** $Y_M = \frac{rK}{4}$ ($E = \frac{r}{2}$) **Steady State:** $N_h | Y_M = \frac{K}{2}$

Continuous Population Model (Y₀): Model: $\frac{dN}{dt} = rN(1 - \frac{N}{K}) - Y_0 = f(N; r, K, Y_0)$ Y_0 : constant yield rate/unit time.

Max Yield: $Y_M = \frac{rK}{4}$ **Equilibrium:** $0 < Y_0 < Y_M$: Two positive points (unstable(小) and stable(大)). **Sensitivity:** 对于 fluctuation 此模型和 EN 模型都不稳定, 但此模型更敏感 sensitive.

Examples of $N_{t+1} = f(N_t)$: Verhulst: $N_{t+1} = rN_t(1 - \frac{N_t}{K})$; $r, K > 0$. Better: **Ricker curve:** $N_{t+1} = N_t e^{r(1 - \frac{N_t}{K})}$

Imperfect Bifurcation: $\dot{x} = rx - x^3 + h$ (codimension-2 bifurcation) If $r < 0$, x^* unique. If $r > 0$, $|h| > h_c$, x^* unique; $|h| < h_c$, x^* three. (两边 stable, 中 unstable). Bifurcation points: $h = \pm h_c = \pm \frac{2\sqrt{r}}{3}$. **cusp point:** $(r, h) = (0, 0)$

1.2 Summary - Stability and Bifurcation

	$\frac{dN}{dt} = f(N)$ No limit cycle periodic solutions S : stable, U : unstable.				$N_{t+1} = N_t F(N_t) = f(N_t)$ Eigenvalue: $\lambda = f'(N^*)$							
Stability	Stable: $f'(N^*) < 0$		Unstable: $f'(N^*) > 0$		Stable: $ \lambda < 1$		Unstable: $ \lambda > 1$ (chaotic)		Periodic: $ \lambda = 1$ $\begin{cases} + : \text{tangent bifurcation} \\ - : \text{period-doubling bifurcation} \end{cases}$ (neutrally stab.)			
Linearise	$N(t) = N^* + n(t), n(t) \ll 1 \Rightarrow \frac{dn}{dt} = \frac{dN}{dt} = \begin{cases} \text{(general)} = f(N^*) + f'(N^*)n(t) \\ \text{(delay)} = \text{直接带入 } t \text{ 后近似} \end{cases}$				Let $N_t = N^* + n_t, n_t \ll 1 \Rightarrow N^* + n_{t+1} = \dots = f(N^* + n_t) \approx f(N^*) + f'(N^*)n_t \Rightarrow n_{t+1} = \lambda n_t$							
Bifurcation \\ Perturbations at N^* Periodic	Saddle-Node (Normal Form) $\dot{x} = r \pm x^2$ $0 \rightarrow S \rightarrow U$ -S	Transcritical $\dot{x} = rx - x^2$ $S \rightarrow U$ -S	(super-)Pitchfork $\dot{x} = rx - x^3$ $S \rightarrow S$ -U-S	Sub-critical Pitchfork $\dot{x} = rx + x^3$ $U \rightarrow U$ -U-S	$\lambda < -1$: $\uparrow$ oscillations		$-1 < \lambda < 0$: $\downarrow$ oscillations		$0 < \lambda < 1$: $\downarrow$ Monotone			
					$\lambda > 1$: $\uparrow$ Monotone							
					If $fP(N^*) = N^*$, N^* is p -periodic. $\quad \forall$ even p -periodic N^* with $\lambda = -1$ branches into $2p$ -periodic							
					Sarkovskii's Theorem: If a 3-periodic N^* exists, then $\forall p \geq 1$, p -periodic N^* exists. Species extinction: $N_{\min} < 1$							
Other	Time-Independent: At N^* : $\frac{dN}{dt} = 0 \quad N(t) = N(t-T) = N^* \quad f(N^*) = 0$				Max/Min: [如果单峰 (unimodal)] 2-periodic $f'(N_{\min}) = 0, N_{\max} = f(N_{\min}); N_{\min} = f(N_{\max}) \downarrow E, Y_0$ 对应两种模型							
Other	Tangency Condition: If $\dot{x} = f(x, r)$, 分岔点 (x^*, r_c) : $\frac{\partial f}{\partial x}(x^*, r_c) = 0 \quad f(x^*, r_c) = 0$ 两点合并: coalesce . Recovery Time: $T = -\frac{1}{\lambda}$ 比收获模型好看 recovery time $\rightarrow$ 算 $\frac{T(E)}{T(0)} \frac{T(Y)}{T(0)}$ 小好											

Hopf Bifurcation: 平衡点从稳定变为不稳定, 同时产生 limit cycle. 条件: $\det(J) > 0$; $\frac{d\text{tr}(J)}{d\mu} \neq 0$ or: $\text{tr}(J)$ $\pm$ -符号有变化 $\Leftrightarrow \lambda = \pm i\omega$ and $\frac{d\Re(\lambda)}{d\mu} \neq 0$ μ : parameter, not time.

$\Rightarrow$ Stable $\rightarrow$ Unstable + Stable Limit Cycle (supercritical) or Unstable $\rightarrow$ Stable + Unstable Limit Cycle (subcritical). (一般 super-) when $\text{tr}(J) = 0 | \Re(\lambda) = 0$, $\lambda = \pm i\omega \Rightarrow$ **periodic:** $T = \frac{2\pi}{\omega}$

Bifurcation: Qualitative changes in dynamics are called *bifurcations*. (当平衡点性质 | 个数发生变化时) **Bifurcation Diagram:** y 轴: 平衡点; x 轴: 控制参数; 虚线 unstable, 实线 stable.

稳定性 Stability 判别 | For Continuous Model: Find perturbation: $n(t)$ and $\frac{dn}{dt} \Rightarrow$ Assume $n(t) = e^{\lambda t} | ce^{\lambda t} \rightarrow$ Find $\lambda = g(\lambda) \Rightarrow$ Let $\lambda = u + i\omega \rightarrow u = g_1(u)$ and $\omega = g_2(\omega) \Rightarrow u < 0$ asy.stable; $u > 0$ unstable. **Find T_c :** (如要寻找参数 (e.g. T) 范围/临界值 使得平衡点稳定) Check $T = 0$ stable. $\Rightarrow$ Using $u = 0$ (i.e. $\lambda = i\omega$) to find T_c .

2 Interacting Populations

Lotka-Volterra Model (Predator-Prey): $\frac{dN}{dt} = N(a - bP)$; $\frac{dP}{dt} = P(cN - d) \Rightarrow$ nondim: $\frac{dN}{d\tau} = u(1 - v)$; $\frac{dP}{d\tau} = \alpha v(u - 1)$ **Equilibrium:** $(u^*, v^*) = (0, 0), (1, 1)$

Exact sol: $\alpha u + v - \ln(u^\alpha v) = H$, $H > H_{\min} = \alpha + 1$ constant. (closed phase) **Conservation Law:** Using $\frac{du}{dv} = \frac{du}{d\tau} \frac{d\tau}{dv}$ find $H \Rightarrow \frac{dH}{d\tau} = 0$, here it satisfies. (By chain rule)

Show Level curves (constant of motion) closed: If $H(u, v)$ is level curve, Look $\begin{bmatrix} H_{uu} & H_{uv} \\ H_{vu} & H_{vv} \end{bmatrix}_{(u^*, v^*)} \Rightarrow$ If $\pm$ - definite (i.e. $\forall \lambda_i > 0 (< 0)$) $\uparrow$ 轨迹会在 level curve $\rightarrow$ oscillation

$\Rightarrow$ It's local min/max at (u^*, v^*) & strictly convex/concave $\Rightarrow$ closed curves. Closed curves $\Rightarrow$ periodic sol.

Determinant Stability: For model: $\begin{bmatrix} u'(t) \\ v'(t) \end{bmatrix} = \begin{bmatrix} f(u, v) \\ g(u, v) \end{bmatrix}$ at equilibrium (u^*, v^*) . Let $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u - u^* \\ v - v^* \end{bmatrix}$, then $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} f_u & f_v \\ g_u & g_v \end{bmatrix}_{(u^*, v^*)} \begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow$ Find eigenvalues λ of J by $\det(J - \lambda I) = 0$

¹ $\det(J) > 0$, $\text{tr}(J) < 0 \Rightarrow \Re(\lambda) < 0 \rightarrow$ stable node $\text{Im}(\lambda) = 0$; stable focus=spiral point $\text{Im}(\lambda) \neq 0$; ² $\det(J) > 0$, $\text{tr}(J) > 0 \Rightarrow \Re(\lambda) > 0 \rightarrow$ unstable node $\text{Im}(\lambda) = 0$; unstable focus=spiral point $\text{Im}(\lambda) \neq 0$

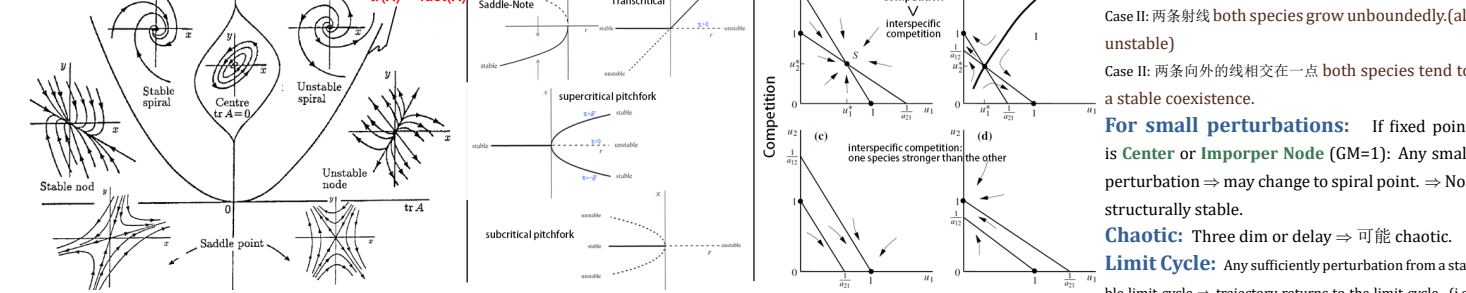
³ $\det(J) < 0 \Leftrightarrow \text{Im}(\lambda) = 0$, $\lambda_1 < 0 < \lambda_2 \rightarrow$ saddle point (unstable); ⁴ $\det(J) > 0$, $\text{tr}(J) = 0 \Rightarrow \lambda = \pm i\omega$ (pure imaginary) $\rightarrow$ center (only linear stability) | Hopf bifurcation (见前面)

Nullclines: $f(u, v) = 0$ or $g(u, v) = 0 \rightarrow$ 得 u, v 表达式是 nullclines. **Trapping Region:** A closed region R 可以用直线/null clines 构成 s.t. 所有边界的外法线方向 $\mathbf{n}$, 在对应边界上的点, 满足 $\mathbf{n} \cdot \begin{bmatrix} f(u, v) \\ g(u, v) \end{bmatrix} < 0$

Find slope of Nullclines: $f(u, v) = 0 \Rightarrow df = f_u du + f_v dv = 0 \Rightarrow \frac{dv}{du} = -\frac{f_u}{f_v}$ Similarly: $\frac{du}{dv} = -\frac{g_u}{g_v}$ $\uparrow$ 也叫 bounding region (confined region)

Poincare-Bendixson Theorem: One unstable node/focus/spiral point + it's in *trapping region* $\Rightarrow$ A **stable limit cycle** exists. (Cannot be a saddle (since $\det(J)$ cannot be < 0))

Competition Model: principal of competitive exclusion: If two species compete for the same limited resource, one will eventually outcompete and exclude the other. (b,c,d; b 物种谁最后存活取决于初始值)



3 Reaction Kinetics

Enzyme Kinetics: e.g. $S + E \xrightleftharpoons[k_{-1}]{k_1} SE \xrightarrow{k_2} P + E$ K_i : rate of reaction; S : substrate; E : enzyme; SE : complex; P : product. **Rate of uptake:** $v = \frac{d[P]}{dt} |_{t=0}$

Law of Mass Action: Rate of reaction is proportional to the product of the concentrations of the reactants. e.g. $\alpha_1 A + \alpha_2 B \xrightleftharpoons[k_{-1}]{k_1} \beta_1 C + \beta_2 D \Rightarrow$ rateforward $= k_1 [A]^{\alpha_1} [B]^{\alpha_2}$; ratebackward $= k_{-1} [C]^{\beta_1} [D]^{\beta_2}$

Quasi-Steady State Approximation (QSSA): 如果系统里有快和慢变量, 则可以假设快变量达到稳态, 即 $\frac{d(\text{fast})}{dt} = 0$. (or $\varepsilon \rightarrow 0$; 一般来说假设 complex concentration changes rapidly)

Singular Perturbations Analysis: For $\frac{du}{d\tau} = f(u, v)$, $\varepsilon \frac{dv}{d\tau} = g(u, v)$, $0 < \varepsilon \ll 1 \Rightarrow v$ fast (when $g \neq 0$), u slow. **Quasi-steady-state approximation:** $\varepsilon \rightarrow 0 \Rightarrow g(u, v) = 0$

Taylor: $u(\tau, \varepsilon) = \sum_{n=0}^{\infty} \varepsilon^n u_n(\tau)$; $v(\tau, \varepsilon) = \sum_{n=0}^{\infty} \varepsilon^n v_n(\tau)$ **Outer solution:** Taylor 带入原方程组 $\rightarrow$ 保留 $O(1) | \varepsilon = 0$ Valid for $\tau > 0$ (slow time); $v_0(0)$ **invalid**. $\rightarrow$ not uniformly valid for $\tau \geq 0$

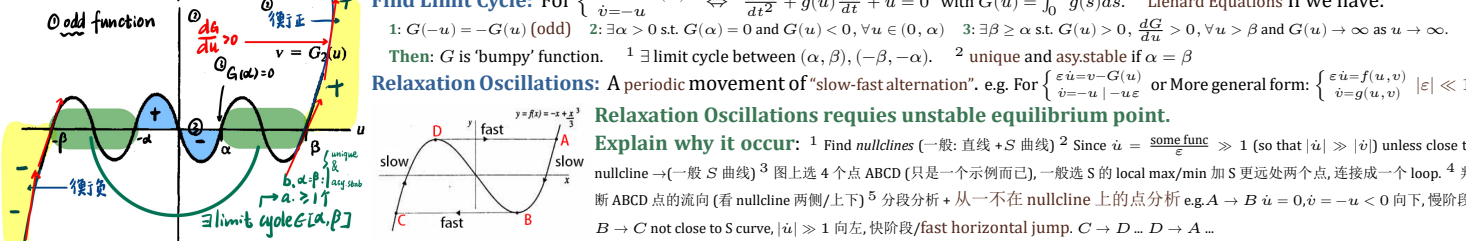
Inner solution: $\sigma = \frac{\tau}{\varepsilon}$, $U(\sigma; \varepsilon) = u(\tau; \varepsilon) \Rightarrow \frac{dU}{d\sigma} = \varepsilon f(U, v)$, $\frac{dV}{d\sigma} = g(U, v) \rightarrow$ Taylor 带入原方程组 $\rightarrow$ 保留 $O(1) | \varepsilon = 0$ (known as **boundary layer** ($\tau = 0$ 处)); Valid $\tau = 0$ (fast time)

Asymptotic Matching: $\lim_{\sigma \rightarrow 0^+} [U_0(\sigma), V_0(\sigma)] = \lim_{\tau \rightarrow 0^+} [u_0(\tau), v_0(\tau)]$ **Uniformly Valid Solution:** $u(\tau; \varepsilon) = u_0(\tau) + O(\varepsilon)$; $v(\tau; \varepsilon) = \begin{cases} v_0(\tau) + O(\varepsilon), & 0 < \varepsilon \ll \tau \\ V_0(\sigma) + O(\varepsilon), & 0 < \tau \ll 1 \end{cases}$ **uniform expansion:** 见 chp.5

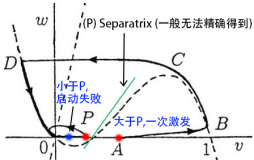
Activator-Inhibitor Model: For $\frac{du}{dt} = f(u, v)$, $\frac{dv}{dt} = g(u, v)$: If $\frac{\partial g}{\partial v} > 0$, u is an **activator** of v ; If $\frac{\partial f}{\partial v} < 0$, v is an **inhibitor** of u . ps: Possible have bifurcation behaviour

Positive/Negative Feedback: At Equilibrium: **Positive:** $f_u, f_v \geq 0$; $g_u, g_v \leq 0$ or $f_u, f_v \leq 0$; $g_u, g_v \geq 0$ **Negative:** $f_u, f_v \geq 0$; $g_u, g_v \geq 0$ or $f_u, f_v \leq 0$; $g_u, g_v \leq 0$

4 Relaxation Oscillations & Switch Behaviour



Estimate period T : 例如上图 (例子忽略了快反应) $T = \int_{t_A}^{t_B} dt + \int_{t_C}^{t_D} dt = 2 \int_{t_A}^{t_B} dt$ Find $dt = _du$ 带入计算 e.g. In slow time: $\dot{u} = 0 \rightarrow v = G(u)$ and $\dot{v} = -u \rightarrow \frac{dv}{dt} = \frac{dv}{du} \frac{du}{dt} = G'(u) \frac{du}{dt} = -u \rightarrow -\frac{G'(u)}{u} du = dt$, 带入计算



Excitable System: FitzHugh-Nagumo Model $\begin{cases} \frac{dx}{dt} = c[y + x - \frac{x^3}{3} + z(t)] \\ \frac{dy}{dt} = \frac{-x + a + by}{c} \end{cases}$ (x : excitability variable; y : recovery variable; $z(t)$: external stimulus)
Threshold 现象: **Small perturbation**: return to equilibrium (failed initiation/excitation). **Large perturbation**: large excursion before returning to equili. (action potential|successful initiation). After excitation|大扰动之后: Refractory Period 不应期: system is unresponsive to further stimulation. **constant stimulus**: periodic firing (limit cycle).
Reason: 寻找 $\dot{u}, \dot{v}$ s.t. $\ll, \gg 1$ i.e. $|\dot{u}| \gg |\dot{v}|$ or $|\dot{u}| \ll |\dot{v}|$
Switch Behaviour: 从一个 stable point 到另一个 stable point. (前面 excitable system|relaxation oscillation 如两有 stable points 且画图分析 (像 relaxation oscillation 那样分析) 可得到)

描述话语 example: 快-慢震荡: ¹ (option) The equilibrium point S is unstable between the two knees/turning points of the u -nullcline/cubic-like curve. | or: S changes stability as it crosses the knees of the u -nullcline. ² 图上标/简单解释拐点位置. ³ 证明 $|u'| \gg 1$ (写成乘积形式判断)或其他 $\rightarrow$ 得快运动边. ⁴ Trajectories starting off the u -nullcline rapidly/quickly approach left/right branch of the u -nullcline, and then slowly move down/up|crawl along the u -nullcline due to $v' < 0 / > 0$, and to upper/lower knee/turning point ⁵ At the knee/turning point, the trajectory quickly jumps to the right/left branch of the u -nullcline, and then slowly moves up/down|crawl along the u -nullcline due to $v' > 0 / < 0$. ⁶ And this process repeats periodically, giving rise to a relaxation oscillation. Excitable: 类似的, 分析 small perturbation above/below equilibrium point $S \rightarrow$ 得 failed excitation or successful excitation.

5 Manifolds & Fenichel's Theorem | SIR Model

Regular/Singular Perturbation: 对于一个含 ε 的方程/系统: If $\varepsilon \rightarrow 0$ the order does not change, it's a **regular perturbation**. If $\varepsilon \rightarrow 0$ the order changes, it's a **singular perturbation**.
Find Roots: Assume roots $x = x_0 + \varepsilon x_1 + \varepsilon^2 x_2 + \dots$, 代入比较 $\mathcal{O}(1), \mathcal{O}(\varepsilon), \dots$ 得系数(解前两系数就行). If singular, cannot approx $x = \mathcal{O}(\varepsilon^{-1})$, rescaling $x = \frac{X}{\varepsilon}$, assume root $X = X_0 + \varepsilon X_1 + \dots$ 类似前. 不留 $X_0 = 0$ 解, 因为 X 小.
Sol & Uniform: For $\varepsilon \dot{y} + p(x)\dot{y} + q(x)y = r(x)$, a. 类似上面算 sol (有初值) or b. 直接算 exact sol. $\rightarrow$ 得 **Outer Solution** (首项含 ε). Let $\tau = \varepsilon t$ 类似算 sol $\rightarrow$ **Inner Solution** (首项不含 ε). **unif expr** = Outer + Inner - Matching(=matching cond (lim))

Singular Perturbation System: For $\begin{cases} \dot{u} = f(u, v, \varepsilon) \\ \varepsilon \dot{v} = g(u, v, \varepsilon) \end{cases}, 0 < \varepsilon \ll 1. \quad \text{If } \tau = \varepsilon t, \text{ then } \begin{cases} \varepsilon u' = f(u, v, \varepsilon) \\ v' = g(u, v, \varepsilon) \end{cases} \quad \begin{matrix} t: \text{fast time}; \tau: \text{slow time} \\ (\varepsilon \neq 0 \text{ 时等价}) \end{matrix} \quad \downarrow \text{Defined in a neighborhood of } \mathcal{M}_0$
Reduced Problem: slow times $\varepsilon = 0$. **Layer Problem**: fast time $\varepsilon = 0$. **Critical Manifold**: $\mathcal{M}_0 := \{(u, v) : f(u, v, 0) = 0\}$ ps: Reduced problem defined only on $\mathcal{M}_0$, Layer problem defined off $\mathcal{M}_0$
Normal Hyperbolicity: $\mathcal{M}_0$ is normally hyperbolic at $(u, v) \in \mathcal{M}_0$ if: λ_i (特征值) of Jacobian $J|_{(u, v, 0)} = \frac{\partial f}{\partial u}|_{(u, v, 0)}$ are $\Re(\lambda_i) \neq 0 \rightarrow$ **(Un-)Stable**: $\Re(\lambda_i) < 0 (> 0) \Leftrightarrow \frac{\partial f}{\partial u} < 0 (> 0)$ (1-dim)
Outer Solution: sol for slow time at $\varepsilon = 0$. (reduced problem) **Inner Solution**: sol for fast time at $\varepsilon = 0$. (layer problem) $\uparrow$ attracting (stable) / repelling (unstable)

Describe Reduced|Layer Problem: ¹ 如果能解具体表达式就解; 不好解情况下给出 u', v' 仅关于 u, v 的表达式; ² 如果可以画一个图; ³ 分析 attracting/repelling (stable/unstable); ⁴ 分析 $\tau, t \rightarrow \infty$ 时的行为
Fenichel's Theorem: If $\mathcal{M}_0 \subset \{f(u, v, 0) = 0\}$ is compact (closed & bounded) and normally hyperbolic. Suppose f, g are smooth. Then for $\varepsilon > 0$ (sufficiently small), $\exists$ **Manifold** $\mathcal{M}_\varepsilon$ s.t. $\mathcal{O}(\varepsilon)$ close and diffeomorphic to $\mathcal{M}_0$. (i.e. $\mathcal{M}_\varepsilon$ is locally invariant under the flow of the full system (上面左边的 system)). ps: $\mathcal{M}_\varepsilon$ called **slow manifold**.

Find $\mathcal{M}_\varepsilon$: If $\mathcal{M}_0 = \{(u, v) : u = p_0(v)\}$, then $\mathcal{M}_\varepsilon = \{(u, v) : u = p_\varepsilon(v)\}$ 可通过算 u, v 的 exact sol, 寻找能使含 ε 项消失但保留 τ 的解 (or: 考虑快变量 $t \rightarrow \infty$), 对比 $\mathcal{M}_0$ 得到 $\mathcal{M}_\varepsilon$
Invariance of Manifold: For slow $\mathcal{M}_\varepsilon|_{\mathcal{M}_0} \rightarrow$ manifold 中的等式关系 $u = p(v)$ 两边对 $\tau|t$ 求导 $\rightarrow$ 代入 slow/fast 对应的 u', v' ; 代入 $\mathcal{M}_\varepsilon|_{\mathcal{M}_0}$ 中的等式关系 $\rightarrow$ 判断是否 LHS=RHS.

SIR: S: # people who are susceptible. (易感人群) I: # people who are infected. (感染者) R: # people who have been infected and recovered. (no longer susceptible). (康复者) **Total Pop**: $N = S + I + R$
Model: $S' = -\beta SI, I' = \beta SI - \alpha I, R' = \alpha I, S, I \geq 0$. **Assumptions**: ¹ Mass Action incidence: βN transmit infection/unit time; ² αI recover/unit time; ³ No im/emigration, births, deaths. ⁴ No deaths from disease. $\rightarrow$ 新感染人数 $= \beta SI$ rate.
临界值: if $S > \frac{N}{\beta}, I \uparrow$ 疾病爆发; if $S < \frac{N}{\beta}, I \downarrow$ 疾病消退. $\Rightarrow$ **Basic Reproduction Number**: (A threshold quantity) $R_0 = \frac{\beta S_0}{\alpha}$ or $\frac{\beta N}{\alpha}, R_0 > 1$ epidemic occurs; $R_0 < 1$ disease dies out.
 $I_\infty = \lim_{t \rightarrow \infty} I(t) = 0; S_\infty = (S + I)_\infty > 0; N = S_0 + I_0$ **Final Size Relation**: $\ln(\frac{S_\infty}{S_0}) = \frac{\beta}{\alpha} [N - S_\infty] = R_0 [1 - \frac{S_\infty}{N}]$ As $(S + I)' = -\alpha I \rightarrow N - S_\infty = \alpha \int_0^\infty I dt$; As $\frac{S'}{S} = -\beta I \rightarrow \ln(\frac{S_\infty}{S_0}) = \beta \int_0^\infty I dt$
Attack Rate/Ratio: $N - S_\infty$ (# people ever infected | 总感染人数 | Final size of epidemic) **Orbits Sol**: $I(t) + S(t) - \frac{\alpha}{\beta} \log(S(t)) = N - \frac{\alpha}{\beta} \log(S_0)$ **Peak**: $I_{\max}$ when $S = \frac{\alpha}{\beta} \Leftrightarrow I' = 0$ (代入前面得精确 $I_{\max}$)
Infective Period: $= \frac{1}{\alpha}$ 就减去的那一项的常数倒数为特征时间尺度 ps: e.g. 可以由此计算单位时间内新感染人数 (从开始): $\int_0^{\alpha^{-1}} I' dt = I(\alpha^{-1}) - I(0)$

6 Reaction Diffusion

Diffusion: Net movement from high to low concentration through random, small-scale motion. **Fickian Diffusion**: $J = -D \frac{\partial c}{\partial x}$ (J : flux; D : diffusivity; c : concentration; $-$: from high to low)
Conservation Relation: $\frac{\partial}{\partial t} \int_{x_0}^{x_1} c(x, t) dx = J(x_0, t) - J(x_1, t)$ $x_1 = x_0 + \Delta x$ When $\Delta x \rightarrow 0 \rightarrow \frac{\partial c}{\partial t} = -\frac{\partial J}{\partial x} = \frac{\partial(D \frac{\partial c}{\partial x})}{\partial x}$ (If D const. Heat Equation)
Solve PDE: For $u(x, t), \mathbf{x} = (x_1, \dots, x_n); u_t = D \Delta u$ with $u(x, 0) = f(x)$ If $f(x) = Q \delta(x)$, then **Fundamental Sol**: $u(x, t) = \phi(x, t) = \frac{Q}{(4\pi Dt)^{n/2}} e^{-\frac{|x|^2}{4Dt}}$ Else: **General Sol**: $u(\mathbf{x}, t) = \int_{\mathbb{R}^n} \phi(\mathbf{x} - \mathbf{y}, t) f(\mathbf{y}) d\mathbf{y}$
Moreover, for $u(\mathbf{x}, t), u_t = D \Delta u + a(t)u$ with $u(\mathbf{x}, 0) = f(\mathbf{x}) \Rightarrow u(\mathbf{x}, t) = v(\mathbf{x}, t) e^{\int_0^t a(s) ds}$ with $v(\mathbf{x}, 0) = f(\mathbf{x})$ satisfying $v_t = D \Delta v$. $\rightarrow$ Use above. **invades const. radial speed**: 算 $\frac{dr}{dt}$
Boundary: ¹ **Diffusion Model**: 认为 $u(x, t)$ 是 bounded (travelling wave 不一定) (含 $\pm\infty$, 极坐标 $r \rightarrow \infty$, 得 IC) ² 对 u, J (flux), 有 continuous. (得 BC, 尤其是分段函数) ³ 对 J (flux), 考虑由物理意义, 边界是否为 0
假设和默认: ¹ 若要假设 $u(x, t)$ 的 IC, 考虑 $u(x, 0) = b_0 > 0$ (均匀的分布) ² **Stationary|Steady-state|Static**: $u_t = 0$. ³ **Homogeneous Rest State|Const. Equilibrium|Uniform Steady State**: $u_x = 0$ and $u_t = 0$.
Useful Tools: ¹ 最后, 可通过两边对 u 对 x 积分得待定系数值 ² **Linearise at u^*** : Let $u = u^* + \eta, |\eta| \ll 1$ func. ³ 对于求解 PDE, 考虑直接计算 or Separation of var. ⁴ **Desingularise**: e.g. For $u(x, t)$, Let $\tau = u t$

7 Travelling Waves

Travelling Wave: $u(x, t) = u(x - ct) = U(z)$ with $z = x - ct$. c : wave speed; z : wave variable If $z = x - ct$ constant, then u constant. **Equilibrium**: 对于时间的求导为 0, $u_t = 0$
Condition: For travelling wave under reaction-diffusion|wavefront, have $u(\pm\infty, t) = u^\pm$, and $u_x(\pm\infty, t) = 0$ or $\lim_{z \rightarrow \pm\infty} U'(z) = \lim_{z \rightarrow \pm\infty} V(z) = 0$ **Spatially Homogeneous**: $u_x = 0$
If $u(x, t) = U(z), z = x - ct$ and $u_t = f(u) + Du_{xx} \Rightarrow$ Change to ODE: $-cU' = f(U) + DU'' \Rightarrow$ By using $V = U' \rightarrow$ ODE System: $\begin{cases} U' = V \\ V' = -\frac{c}{D} V - \frac{f(U)}{D} \end{cases}$
For $\uparrow$ travelling wave **boundary conditions**: $U(-\infty) = u^-$ (ahead of wave); $U(\infty) = u^+$ (behind wave) **Heteroclinic orbit**: A trajectory connecting $(u^-, 0)$ and $(u^+, 0)$ (**Homoclinic Orbit**: $u^- = u^+$)
证明异宿轨道存在: Using $\frac{dU}{dV} = \frac{U'}{V'}$ to find $V(U)$. If $V(u^\pm) = 0$, then exists. **寻找波速 c** : Find ODE: $D U'' + c U' + f(U) = 0 \rightarrow$ 乘 U' 两边积分: $D \int_{-\infty}^{\infty} U'' U' dz + c \int_{-\infty}^{\infty} (U')^2 dz + \int_{-\infty}^{\infty} f(U) U' dz = 0$
 $\uparrow$ show: reasonable exist u s.t. $\lim_{z \rightarrow \pm\infty} u(z) = u^\pm$ exists || $\rightarrow$ using $\int_{-\infty}^{\infty} U'' U' dz = \frac{1}{2} [(U')^2]_{-\infty}^{\infty} \rightarrow \int_{-\infty}^{\infty} U'' U' dz = 0$ and $U' dz = dU \rightarrow c = \frac{\int_{u^-}^{u^+} f(U) dU}{\int_{-\infty}^{\infty} (U')^2 dz}$ (ps: c 符号 面积差决定)
Find equilibrium: $V = 0$ and $U_1, U_2, U_3 \rightarrow$ 计算 Jacobian + 判断 stability $\rightarrow$ 边上两 saddle, 中间 node/focus $\rightarrow$ only interest in the unstable eigendirection $\rightarrow$ 不用算/简短算出 U_1, U_3 处的 eigenvector
方向: 说明 ¹ 一个方向进 (stable eigendirection, 特征值 < 0), 一个方向出 (unstable eigendirection, 特征值 > 0); ² 左边进/出的方向 (positive/negative slope), 右边出/进的方向 (negative/positive slope). i.e. 有

交点 + 一进一出 $\Rightarrow$ possible for a heteroclinic connection between U_1 and U_3 . || if $c = \frac{\int_{u^-}^{u^+} f(U) dU}{\int_{-\infty}^{\infty} (U')^2 dz} = 0$, **critical front speed** 对于灾害: $c < 0$ no spread; $c > 0$ outbreak spread

Fisher-Kolmogoroff: Eq: $u_t = Du_{xx} + ru(1 - \frac{u}{K}) \Leftrightarrow U'' + \frac{c}{D} U' + \frac{r}{D} U(1 - \frac{U}{K}) = 0$ when $u(x, t) = U(z), z = x - ct$. 如有 **wavefront sol**: **Wave Speed**: $c \geq c_{\min} = 2$, and $\lim_{z \rightarrow -\infty} U(z) = K(0)$.

Remark: 1. 对于单 PDE/ODE 中, Equilibrium 的稳定性判断看 $f'(u^*)$ 符号; 对于 ODE System 中, 看 Jacobian. (两者不同) 2. $u(x, t)$ 如用 Separation of var. 算出 $x = \text{const}$. 要 wave sol 时忽略.

For $\lambda = 0$ Case|Degenerate: 判 Stability 时如特征值为: $\lambda_1 = 0, \lambda_2 < 0$: It's a "saddle-node" with one stable eigendirection (corresponding to $\lambda_2 < 0$) and one "center" eigendirection (corresponding to $\lambda_1 = 0$).

8 Appendix

$y' + p(x)y = q(x)$: **Integrating Factor**: $I(x) = e^{\int p(x) dx} \Rightarrow I(x)y = \int I(x)q(x) dx$ **Taylor Expansion**: $f(x, r)$ at (x_0, r_0) : $f(x, r) = f(x_0, r_0) + (x - x_0)f_x|_{(x_0, r_0)} + (r - r_0)f_r|_{(x_0, r_0)} + \frac{1}{2} \dots$
Binomial Series: $(1 + x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!} x^2 + \dots$ **Linearisation**: $f(x + \Delta x) = f(x) + f'(x)\Delta x + \dots$ $e^{bx} = 1 + bx + \frac{(bx)^2}{2!} + \dots$ $\ln(a + x) = \ln(a) + \frac{x}{a} - \frac{x^2}{2a^2} + \frac{x^3}{3a^3} - \dots$ $y(a) - y(b) = \int_b^a y'(x) dx$
Geometry Series: $\frac{1}{1-x} = 1 + x + x^2 + \dots, |x| < 1$ **Hyperbolic Funcs**: $\sinh x = \frac{e^x - e^{-x}}{2}$ $\cosh x = \frac{e^x + e^{-x}}{2}$ $\tanh x = \frac{\sinh x}{\cosh x}$ $\sinh(x) = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots$ $\cosh(x) = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$ $\tanh(x) = x - \frac{x^3}{3} + \frac{2x^5}{15} - \dots$
Integral: $\arctanh(x) = \frac{1}{2} \ln(\frac{1+x}{1-x})$ $\text{arcsinh}(x) = \ln(x + \sqrt{x^2 + 1})$ $\text{arccosh}(x) = \ln(x + \sqrt{x^2 - 1})$ $\int \frac{1}{\sqrt{a^2 - x^2}} dx = \arcsin(\frac{x}{a})$ $\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \arctan(\frac{x}{a})$
Useful Tools: $\Delta u = u_{xx} + u_{yy} = u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta}$ If u **radial symmetry**: $u = u(r, t)$ (i.e. $u_\theta = 0$) $dA = r dr d\theta$ in polar coord. and $dV = r^2 \sin \phi dr d\theta d\phi$ in spherical coord.
 $ay'' + by' + cy = 0; y'' - \lambda y = 0$: ¹ $\lambda > 0$: $y = C_1 \cosh(\sqrt{\lambda}x) + C_2 \sinh(\sqrt{\lambda}x) = A e^{\sqrt{\lambda}x} + B e^{-\sqrt{\lambda}x}$; ² $\lambda = 0$: $y = C_1 + C_2 x$; ³ $\lambda < 0$: $y = C_1 \cos(\sqrt{-\lambda}x) + C_2 \sin(\sqrt{-\lambda}x)$
General: characteristic eq: $ar^2 + br + c = 0$, root $r_{1,2} = \mu \pm \nu i$. ¹ $\Delta > 0$: $y = C_1 e^{r_1 x} + C_2 e^{r_2 x}$; ² $\Delta = 0$: $y = e^{\mu x} (C_1 + C_2 x)$; ³ $\Delta < 0$: $y = e^{\mu x} (C_1 \cos(\nu x) + C_2 \sin(\nu x))$
Separation of var: ¹ 假设 $u(x, t) = X(x)T(t)$ $\rightarrow$ 带入 PDE ² 解出类似 $\frac{T'}{T} = \frac{X''}{X} = -\lambda$ $\rightarrow$ 得到两个 ODE ³ 解较为简单的 ODE, 代入 BC 解 λ and sol. (注意看是否只要 periodic; 极坐标有 $r \rightarrow \infty$ bounded, $\theta = \theta + 2\pi$ 等条件). ⁴ 解另外一个. ⁵ 叠加所有 λ_n 的 sol 得到 PDE 的 general sol $u = \sum c_n X_n(x) T_n(t)$ ⁶ 用 IC 解出 c_n . Remark: ¹ 前期可以用正比化简; ² λ 为让 sin/cos 解为正好计算.

Polar Coord: $x = r \cos \theta, y = r \sin \theta \rightarrow 2rr' = 2xx' + 2yy'$ and $r^2 \theta' = xy' - yx'$